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FORGING AHEAD



You've made it through the dark forest of searching and sorting, across the frozen river of esoteric mathematics, over the treacherous mountain passes of gradient ascent, past the swamp of geometric despair, and you've conquered the dragon of slow runtimes. Congratulations. If you wish, you're free to return to your comfortable home in a land free from algorithms. This chapter is for those who instead wish to continue the adventure after they close this book.

No single book can contain everything about algorithms. There is too much to know, and more is being discovered all the time. This chapter is about three things: doing more with algorithms, using them in better and faster ways, and solving their deepest mysteries.

In this chapter, we'll build a simple chatbot that can talk to us about previous chapters of the book. Then we'll discuss some of the hardest problems in the world and how we might make progress toward crafting